Splunk SPL Investigation Queries

04 - Credential Attack Investigation

Search 1: LSASS Memory Access & Triage Analysis

Command (Splunk SPL)

```
index="soc-investigation" source="*credential_attack*" EventCode="10" TargetProcessName="lsass.exe"
| dedup _time, ComputerName, ProcessId, GrantedAccess
| eval Access_Risk=case(
    GrantedAccess="0x1fffff", "1. CRITICAL - Full Memory Dump Access (0x1fffff)",
    ProcessName="Taskmgr.exe", "2. Review - Task Manager Query (0x1000)",
    1=1, "3. Benign - System / Defender Scan"
  )
| table _time, ComputerName, User, ParentProcessName, ProcessName, ProcessId, TargetProcessName, GrantedAc
cess, Access_Risk
| sort Access_Risk, _time
```

Explanation

• Queries Sysmon EventCode 10 (Process Access) where TargetProcessName is lsass.exe to inspect handles opened to Windows authentication memory.

• Access_Risk evaluates the GrantedAccess rights mask, classifying 0x1fffff (PROCESS_ALL_ACCESS) as CRITICAL malicious execution while separating benign Defender and Task Manager operations.

• dedup removes duplicate log entries across identical access instances, while table and sort organize the findings chronologically by severity tier.

Search 2: Credential Dumping LOLBin Execution Isolation

Command (Splunk SPL)

```
index="soc-investigation" source="*credential_attack*" EventCode="4688" ComputerName="SRV-APP-02" User="SR
V-APP-02\admin_backup" ProcessName IN ("rundll32.exe", "reg.exe")
| dedup _time, ProcessName, CommandLine
| table _time, ComputerName, User, ProcessName, ProcessId, CommandLine
| sort _time
```

Explanation

• Isolates Process Creation (EventCode 4688) events on host SRV-APP-02 executed under the compromised admin_backup user context.

• Filters specifically for native Living-off-the-Land Binaries (LOLBins): rundll32.exe invoking comsvcs.dll and reg.exe exporting registry hives.

• Extracts command-line arguments demonstrating attempts to create debug_crash.dmp alongside sam.hiv, sys.hiv, and sec.hiv in C:\Windows\Temp\.

Search 3: Parent-Child Process Ancestry & Multi-Event Correlation

Command (Splunk SPL)

```
index="soc-investigation" source="*credential_attack*" ComputerName="SRV-APP-02" User="SRV-APP-02\admin_ba
ckup"
| dedup _time, EventCode, ProcessName
| eval Action=case(
    EventCode="4688", "1 - Command Executed",
    EventCode="11", "2 - File Created",
    EventCode="3", "3 - Network Connection",
    EventCode="10", "4 - Memory Accessed"
  )
| eval Evidence=coalesce(CommandLine, TargetFilename, DestinationIp . ":" . DestinationPort, TargetProcess
Name)
| table _time, Action, ProcessName, Evidence
| sort _time
```

Explanation

• Correlates telemetry across Process Creation (4688), File Creation (11), Network Connections (3), and Process Access (10).

• coalesce() evaluates and selects the first non-null artifact from command strings, created file paths, destination sockets, or target process names.

• Forms a chronological timeline tracking the full attack lifecycle: staging (makecab.exe), exfiltration (curl.exe to 198.51.100.77:8443), and evidence cleanup (cmd.exe /c del).

Search 4 — Panel 1: Malicious Process Activity (Pie Chart)

Command (Splunk SPL)

```
/* Panel 1: Malicious Process Activity Distribution Pie Chart */
index="soc-investigation" source="*credential_attack*" EventCode="4688" ComputerName="SRV-APP-02" User="SR
V-APP-02\admin_backup" ProcessName IN ("rundll32.exe", "reg.exe", "makecab.exe", "curl.exe", "cmd.exe")
| dedup _time, ProcessName, CommandLine
| stats count by ProcessName
```

Explanation

• Identifies all binaries leveraged by admin_backup on SRV-APP-02 during the attack.

• stats count by ProcessName aggregates execution frequencies for rundll32.exe, reg.exe, makecab.exe, curl.exe, and cmd.exe to render the breakdown pie chart.

Search 5 — Panel 2: Critical LSASS Memory Access (Table Panel)

Command (Splunk SPL)

```
/* Panel 2: Critical LSASS Memory Access Table */
index="soc-investigation" source="*credential_attack*" EventCode="10" TargetProcessName="lsass.exe" Grante
dAccess="0x1fffff"
| dedup _time, ProcessId
| table _time, ComputerName, User, ProcessName, TargetProcessName, GrantedAccess
| rename _time AS "Timestamp", ComputerName AS "Host", User AS "Compromised Account", ProcessName AS "Atta
cker Process", GrantedAccess AS "Access Level"
| fieldformat Timestamp=strftime(Timestamp, "%Y-%m-%d %H:%M:%S")
```

Explanation

• Filters Sysmon EventCode 10 for handles opened to lsass.exe with full memory access permissions (0x1fffff).

• Uses rename to establish clean column headers suitable for SOC incident reporting.

• fieldformat converts raw Epoch timestamps into standard date-time representations without altering underlying data types.

Search 6 — Panel 3: Credential Theft Commands (Table Panel)

Command (Splunk SPL)

```
/* Panel 3: Credential Dumping Commands Table */
index="soc-investigation" source="*credential_attack*" EventCode="4688" ComputerName="SRV-APP-02" ProcessN
ame IN ("rundll32.exe", "reg.exe")
| dedup _time, CommandLine
| table _time, ProcessName, CommandLine
| rename _time AS "Timestamp", ProcessName AS "Process", CommandLine AS "Malicious Command Executed"
| fieldformat Timestamp=strftime(Timestamp, "%Y-%m-%d %H:%M:%S")
```

Explanation

• Extracts command-line invocations responsible for saving comsvcs.dll memory minidumps and registry hives.

• dedup eliminates identical command lines caused by duplicate dataset indexing.

• Standardizes the Timestamp field into human-readable format for presentation on the dashboard.

Search 7 — Panel 4: Exfiltration Network Traffic (Single Value)

Command (Splunk SPL)

```
/* Panel 4: Exfiltration Network Destination Single Value */
index="soc-investigation" source="*credential_attack*" EventCode="3" ProcessName="curl.exe"
| dedup _time, DestinationIp
| eval Exfil_Target=DestinationIp . ":" . DestinationPort
| table Exfil_Target
```

Explanation

• Queries Network Connection telemetry (EventCode 3) generated by curl.exe.

• Concatenates DestinationIp and DestinationPort into Exfil_Target to highlight the external C2 socket (198.51.100.77:8443).

• Formats the output for display in a high-priority indicator panel.

Search 8 — Panel 5: Post-Exploitation Timeline (Table Panel)

Command (Splunk SPL)

```
/* Panel 5: Post-Exploitation Actions Timeline Table */
index="soc-investigation" source="*credential_attack*" ComputerName="SRV-APP-02" User="SRV-APP-02\admin_ba
ckup" ProcessName IN ("makecab.exe", "curl.exe", "cmd.exe")
| dedup _time, EventCode, ProcessName
| eval Action=case(
    ProcessName="makecab.exe", "1 - Data Staged (.cab)",
    ProcessName="curl.exe", "2 - Data Exfiltrated (Network)",
    ProcessName="cmd.exe", "3 - Evidence Wiped (del)"
  )
| eval Evidence=coalesce(CommandLine, DestinationIp . ":" . DestinationPort)
| table _time, Action, ProcessName, Evidence
| sort _time
```

Explanation

• Isolates post-credential-theft activity involving staging, data transfer, and indicator removal.

• Action labels and enumerates the sequence of actions: archive compression, network egress, and command-line file deletion.

• Uses coalesce to merge command lines and external network endpoints into a unified evidence.